Evaluation Metrics — Complete Guide

Regression + Classification | Formulae + Real-World Examples

> **One dataset, two tasks — used throughout this entire guide:** A hospital has data on 1,000 patients.
>
> - **Regression Task:** Predict a patient's exact **Blood Sugar level** (e.g., 142 mg/dL)
> - **Classification Task:** Predict whether a patient is **Diabetic (1) or Not Diabetic (0)**

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SECTION A — Regression Metrics

> **When to use:** Whenever your model outputs a **continuous number** — house price, temperature, salary, blood sugar, stock price. There is no "correct" or "wrong" class — only how *far off* the prediction was.

A1. Error — The Foundation of Everything

Before any metric, understand the **raw error** for a single prediction:

```
Error  =  Predicted Value  -  Actual Value
       =  y_hat  -  y
```

Example:

Patient	Actual Blood Sugar	Predicted	Error
P1	145 mg/dL	142 mg/dL	−3
P2	160 mg/dL	175 mg/dL	+15
P3	120 mg/dL	118 mg/dL	−2
P4	200 mg/dL	165 mg/dL	−35
P5	95 mg/dL	97 mg/dL	+2

Errors can be **positive** (over-predicted) or **negative** (under-predicted). If you simply average the raw errors, the positives and negatives cancel out — giving a misleading score near zero even when predictions are way off. This is why we need MAE, MSE, and RMSE.

A2. MAE — Mean Absolute Error

What it is

Take the absolute value of each error (remove the sign), then average them.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} |\hat{y}_i - y_i|$$

Calculation

```
|-3| + |+15| + |-2| + |-35| + |+2|
= 3 + 15 + 2 + 35 + 2
= 57

MAE  =  57 / 5  =  11.4 mg/dL
```

> "On average, my model's blood sugar prediction is **off by 11.4 mg/dL**."

The unit of MAE is the **same as the target variable** — making it directly interpretable by a doctor.

Real-World Intuition

A delivery app predicts your order arrives in 30 minutes. It sometimes arrives in 25, sometimes in 40. MAE = 5 minutes means — on average, the app is off by 5 minutes either way. A doctor reading MAE = 11.4 mg/dL immediately understands the practical magnitude of the error.

Properties

Property	Detail
Units	Same as target variable — easy to interpret
Outlier sensitivity	**Low** — treats a 3 mg/dL error and a 35 mg/dL error proportionally
When to use	When all errors are roughly equal importance; don't want big errors to dominate
When NOT to use	When a large error (dangerous misprediction) should be penalised much more heavily

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A3. MSE — Mean Squared Error

What it is

Square each error before averaging. Squaring removes the sign AND magnifies larger errors.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^{n} (\hat{y}_i - y_i)^2$$

Calculation

```
(-3)² + (15)² + (-2)² + (-35)² + (2)²
= 9 + 225 + 4 + 1225 + 4
= 1467

MSE  =  1467 / 5  =  293.4 mg²/dL²
```

> "The average squared error is 293.4."

The unit is now **mg^2/dL^2** — squared units, which are NOT directly interpretable. This is MSE's main weakness.

Real-World Intuition

A self-driving car predicting when to brake. A 1-second error might be fine. A 5-second error could be fatal. MSE punishes the 5-second error **25× more** than the 1-second error (25^2 vs 1^2), not just 5×. This is exactly what you want when large errors are disproportionately dangerous.

Properties

Property	Detail
Units	Squared units — NOT directly interpretable
Outlier sensitivity	**High** — one large error dominates the score heavily
Mathematical advantage	Differentiable everywhere — smooth for Gradient Descent
When to use	When large errors are much more costly than small ones

Property	Detail
When NOT to use	When you want an interpretable, human-readable error metric

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A4. RMSE — Root Mean Squared Error

What it is

Take the square root of MSE. This brings units back to the original scale.

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (\hat{y}_i - y_i)^2}$$

Calculation

```
RMSE  =  sqrt(293.4)  =  17.1 mg/dL
```

> "My model's predictions are off by approximately **17.1 mg/dL**, with larger errors penalised more."

MAE vs RMSE — The Critical Comparison

```
MAE  = 11.4 mg/dL
RMSE = 17.1 mg/dL

RMSE > MAE  →  This always happens when there are large individual errors.
The gap tells you: "There are some really bad predictions hiding in the average."
```

Patient P4 had an error of −35. MAE treats this as just another error. RMSE punishes it hard ($35^2 = 1225$), pulling the score up significantly.

Real-World Intuition

A weather forecasting model usually off by 2°C but occasionally off by 15°C (during storms). The MAE might look decent (say 3°C average), but RMSE will be much higher because of those extreme storm-day errors. RMSE says — "*I care about your worst days, not just your average day.*"

Properties

Property	Detail
Units	Same as target — interpretable like MAE
Outlier sensitivity	High — large errors are penalised quadratically
Best for	When big errors are costly and you want to surface them
Relationship to MAE	RMSE ≥ MAE always. Large gap = outlier errors present.

A5. R^2 — R-Squared (Coefficient of Determination)

What it is

R^2 answers a different question: **"How much of the variation in the target does my model explain?"**

It compares your model against the dumbest possible baseline — a model that predicts the **mean** of y for every single patient.

$$SS_{\text{res}} = \sum_{i=1}^{n} (\hat{y}_i - y_i)^2 \quad \leftarrow \text{your model's total squared error}$$

$$SS_{\text{tot}} = \sum_{i=1}^{n} (\bar{y} - y_i)^2 \quad \leftarrow \text{dumb baseline's total squared error}$$

$$\boxed{R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}}$$

Calculation

```
y_mean = (145 + 160 + 120 + 200 + 95) / 5 = 144

SS_res = 9 + 225 + 4 + 1225 + 4 = 1467        ← your model

SS_tot = (145−144)² + (160−144)² + (120−144)² + (200−144)² + (95−144)²
       = 1 + 256 + 576 + 3136 + 2401
       = 6370                                   ← dumb baseline

R²  =  1  −  (1467 / 6370)  =  1  −  0.23  =  0.77
```

> "My model explains **77% of the variance** in blood sugar levels. The remaining 23% is unexplained noise."

Interpreting R^2 Values

R^2 value	Interpretation
1.0	Perfect — model explains everything
0.9	Excellent
0.7	Good — useful model with some unexplained variation
0.5	Moderate — better than guessing, but misses a lot
0.0	Useless — same as just predicting the mean
< 0	Model is **worse** than predicting the mean — something is very wrong

Real-World Intuition

Predicting a student's exam score using only hours slept: $R^2 \approx 0.15$ — sleep explains only 15% of variation. Add study hours, practice tests, and subject difficulty: R^2 jumps to 0.82. R^2 tells you **how much of the story your features are telling**.

> **Warning:** R^2 can be artificially inflated by adding more features — even useless ones. This is why **Adjusted R^2** exists for multiple regression: it penalises for adding features that don't genuinely help.

A6. Side-by-Side Regression Metrics Summary

Metric	Formula	Unit	Outlier Sensitive?	Interpretable?	Best For
MAE	$\frac{1}{n}\sum\lvert\hat{y}-y\rvert$	Same as y	No	Yes	General purpose, robust to outliers
MSE	$\frac{1}{n}\sum(\hat{y}-y)^2$	Squared units	Yes	No	Gradient Descent optimisation
RMSE	$\sqrt{\text{MSE}}$	Same as y	Yes	Yes	When large errors are costly
R^2	$1 - SS_{\text{res}}/SS_{\text{tot}}$	Unitless (0 to 1)	Moderate	Yes	Understanding explanatory power

> **In practice:** Always report both **RMSE** (to surface big errors) and **R^2** (to understand variance captured). MAE is the most business-friendly metric — non-technical stakeholders instantly understand "off by 11 mg/dL."

SECTION B — Classification Metrics

> **When to use:** Whenever your model outputs a **class label** — Diabetic/Not Diabetic, Spam/Not Spam, Fraud/Legitimate, Cat/Dog. The output is discrete, not continuous.

B1. The Confusion Matrix — Everything Flows From Here

Run your model on 200 test patients. For each patient, compare predicted label vs actual label. Tally into a 2×2 grid:

```
                         PREDICTED
                   Not Diabetic (0)    Diabetic (1)
              ┌─────────────────────────────────────┐
   A  Not     │                      │              │
   C  Diab.   │    TN = 110  ✅       │    FP = 10   │
   T  (0)     │                      │      ❌      │
   U          ├─────────────────────────────────────┤
   A  Diab.   │                      │              │
   L  (1)     │    FN = 11   ❌       │    TP = 69   │
              │                      │      ✅      │
              └─────────────────────────────────────┘
```

The 4 Cells — Memorise These

Cell	Full Name	What happened	Error Type
TP = 69	True Positive	Predicted Diabetic, Actually Diabetic	Correct
TN = 110	True Negative	Predicted Not Diabetic, Actually Not Diabetic	Correct
FP = 10	False Positive	Predicted Diabetic, Actually NOT Diabetic	**Type I Error**
FN = 11	False Negative	Predicted Not Diabetic, Actually WAS Diabetic	**Type II Error**

> **Memory trick:** The first word (True/False) = was the prediction correct? The second word (Positive/Negative) = what did the model predict?

The Cost of Each Error

```
FP (False Alarm):   Doctor calls healthy patient diabetic
                    → Unnecessary tests — inconvenient, costly, stressful
                    → But patient is FINE

FN (Missed Case):   Doctor misses a real diabetic
                    → Patient goes home untreated
                    → Disease progresses → Could be FATAL
```

In medical diagnosis, **FN is the more dangerous error**. Different domains have different cost asymmetries — this shapes which metric you optimise.

B2. Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{69 + 110}{200} = \frac{179}{200} = 89.5\%$$

> "My model correctly classified **89.5% of all patients**."

The Accuracy Trap — When Accuracy Lies

A new dataset: 950 healthy patients, 50 diabetic (heavily imbalanced). A completely useless model predicting **everyone is healthy** scores:

```
Accuracy  =  950 / 1000  =  95%
```

95% accuracy — but it **missed every single diabetic patient**. This is the **accuracy paradox**.

> **Rule:** Only trust Accuracy when classes are roughly balanced. For imbalanced data, always check Precision, Recall, and F1.

B3. Precision

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{69}{69 + 10} = \frac{69}{79} = 87.3\%$$

> "Of all patients **flagged as Diabetic**, **87.3%** were actually diabetic."

Real-World Intuition — Spam Filter

Spam filter flagged 100 emails. Precision = 87% means 87 were actually spam, but 13 were legitimate emails wrongly sent to spam. Precision asks: **"When you cry wolf, how often is it really a wolf?"**

When Precision Matters Most

When **False Positives are costly:** spam filters, drug approval, legal convictions.

B4. Recall (Sensitivity / True Positive Rate)

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{69}{69 + 11} = \frac{69}{80} = 86.3\%$$

> "Of all patients who **actually were Diabetic**, my model caught **86.3%** of them."

The remaining 13.7% (FN = 11) were missed — diabetic but the model said they weren't.

Real-World Intuition — Airport Security

Scanner checks 500 bags. 20 bags contain weapons. Scanner finds 17 (misses 3): Recall = 17/20 = 85%. Recall asks: **"Of all the real threats, how many did you catch?"**

When Recall Matters Most

When **False Negatives are costly:** cancer screening, fraud detection, fire alarms.

> **The fundamental tradeoff:** You can almost always improve Recall by lowering the prediction threshold — flag more patients as diabetic. But this increases FP → Precision drops. This is the **Precision-Recall tradeoff**.

B5. The Precision-Recall Tradeoff

With Logistic Regression, default threshold = 0.5. Moving it changes the balance:

Threshold	Effect	Precision	Recall	FP	FN
0.7 (strict)	Predict Diabetic only when very confident	High	Low	Few	Many
0.5 (default)	Balanced	87.3%	86.3%	10	11
0.3 (lenient)	Predict Diabetic more aggressively	Low	High	Many	Few

```
Strict threshold (0.7):   Miss fewer healthy people (low FP)
                          But miss more real diabetics (high FN)
```

```
Lenient threshold (0.3):  Catch almost every diabetic (low FN)
                          But flag many healthy people too (high FP)
```

> There is no universally "right" threshold — it depends on the **cost of each error type** in your domain. A doctor setting a diabetes screening threshold should weight FN much more heavily than FP.

B6. F1 Score

The Problem it Solves

You need ONE number to compare two models, but Precision and Recall point in different directions. F1 is the **harmonic mean** of both — it punishes extreme imbalances.

$$\text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \times 0.873 \times 0.863}{0.873 + 0.863} = \frac{1.507}{1.736} = 86.8\%$$

Why Harmonic Mean, Not Simple Average?

Arithmetic mean of 90% Precision and 10% Recall = **50%** — misleadingly decent. Harmonic mean (F1) = **18%** — correctly reflects that this model is terrible.

The harmonic mean is always closer to the **smaller** of the two values. A model cannot hide a terrible Recall behind a great Precision.

Comparing Two Models

Model	Precision	Recall	Arithmetic Mean	F1 Score
Model A	90%	30%	60%	**44.4%** — Recall is a serious problem
Model B	75%	75%	75%	**75%** — Balanced

F1 correctly identifies Model B as superior despite Model A having higher precision. The arithmetic mean was fooled.

When to use F1

Imbalanced datasets, single-metric model comparison, NLP tasks (information retrieval, named entity recognition).

B7. Specificity (True Negative Rate)

$$\text{Specificity} = \frac{TN}{TN + FP} = \frac{110}{110 + 10} = \frac{110}{120} = 91.7\%$$

> "Of all patients who **were NOT diabetic**, my model correctly identified **91.7%** as healthy."

Recall vs Specificity

```
Recall (Sensitivity)  →  How well the model catches POSITIVE cases (diabetics)
Specificity           →  How well the model catches NEGATIVE cases (healthy
patients)

Together they give the complete performance picture across BOTH classes.
```

Real-World Intuition — HIV Testing

- **High Recall:** Ensure almost every HIV-positive person is detected — no missed cases
- **High Specificity:** Ensure almost every HIV-negative person is correctly cleared — no false positives causing panic

A good diagnostic test needs both. In practice, increasing sensitivity often reduces specificity — hence the need for confirmatory tests.

B8. ROC Curve and AUC

What is the ROC Curve?

ROC = **Receiver Operating Characteristic**. It plots:

- **Y-axis:** Recall (True Positive Rate) — how many real diabetics are caught
- **X-axis:** False Positive Rate (1 − Specificity) — how many healthy patients are wrongly flagged

As you sweep the threshold from 1.0 down to 0.0, you trace a curve:

```
Recall (TPR)
1.0 |               .─────────── Perfect model
    |           .───'
    |       .──'            ← Your model's curve (AUC = 0.92)
    |    .──'
    | .──'
    |'                      ← Random guessing (AUC = 0.5, diagonal)
    └──────────────────────────
   0.0                     1.0
          FPR (1 – Specificity)
```

AUC — Area Under the ROC Curve

AUC	Interpretation
1.0	Perfect model
0.9+	Excellent
0.8	Good
0.7	Fair

AUC	Interpretation
0.5	Random guessing — no better than a coin flip

> "If I pick one random diabetic patient and one random healthy patient, my model assigns a higher probability of diabetes to the diabetic patient **92% of the time.**"

Why AUC is Powerful for Comparison

AUC is **threshold-independent** — it evaluates the model's inherent discriminating power across ALL possible thresholds at once.

```
Model A (SVM):           AUC = 0.94   ← Best overall discriminator
Model B (Logistic Reg): AUC = 0.91
Model C (KNN, K=7):      AUC = 0.87
```

> **AUC limitation:** On highly imbalanced datasets (1% positive, 99% negative), AUC can look impressive even for a poor model. Use **Precision-Recall AUC** instead for severely imbalanced data.

B9. Log Loss (Cross-Entropy Loss)

What it is

Unlike all other classification metrics that only look at the final label, Log Loss evaluates the **confidence** of predictions — it penalises models that are wrong AND confident.

$$\text{Log Loss} = -\frac{1}{n} \sum_{i=1}^{n} \left[y_i \log(p_i) + (1-y_i)\log(1-p_i) \right]$$

Real-World Intuition

Actual	Model's Predicted Probability	Log Loss Contribution
Diabetic (1)	p = 0.95	Very low — correct AND confident
Diabetic (1)	p = 0.55	Medium — correct but barely confident
Diabetic (1)	p = 0.05	Very HIGH — wrong AND very confident

A model that says "I'm 95% sure this patient is healthy" and is wrong gets **catastrophically penalised**. Log Loss rewards models that are calibrated — confident when right, uncertain when unsure.

```
Perfect model:  Log Loss → 0
Random model:   Log Loss ≈ 0.69
Terrible model: Log Loss → very large
```

When to use

When **probability calibration** matters (risk scoring in medicine or finance), comparing probabilistic models, or training Logistic Regression (Log Loss IS the cost function being minimised).

B10. Complete Classification Metrics Summary

Metric	Formula	What it asks	When to prioritise
Accuracy	$(TP+TN)\ /\ \text{Total}$	Overall, how often is the model correct?	Balanced classes only
Precision	$TP\ /\ (TP+FP)$	When you predict Positive, how often are you right?	FP is costly (spam, false convictions)
Recall	$TP\ /\ (TP+FN)$	Of all real Positives, how many did you catch?	FN is costly (disease, fraud, fire)
F1 Score	$2 \cdot P \cdot R\ /\ (P+R)$	Balanced view of Precision and Recall	Imbalanced data, single-metric comparison
Specificity	$TN\ /\ (TN+FP)$	Of all real Negatives, how many were correctly identified?	Paired with Recall for full picture
AUC-ROC	Area under ROC curve	Overall discriminating power across all thresholds	Comparing models; threshold-independent
Log Loss	$-\text{Mean}[y\log p]$	How confident and correct are your probability outputs?	When calibrated probabilities matter

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B11. Choosing the Right Metric — Decision Guide

```
Is your dataset balanced?
├── YES → Accuracy is a reasonable starting metric
│         Add Precision/Recall for deeper understanding
└── NO  → NEVER use Accuracy alone
          Use F1, AUC-ROC, or Precision-Recall curve

What is more dangerous — FP or FN?
├── FN is more dangerous (disease, fraud, fire, security)
│    → Optimise RECALL → Lower prediction threshold
├── FP is more dangerous (spam filter, legal, drug approval)
│    → Optimise PRECISION → Raise prediction threshold
└── Both matter equally
     → Optimise F1 Score

Do you need to compare models holistically?
└── Use AUC-ROC (threshold-independent, single clean comparison)

Do you care about probability outputs (not just labels)?
└── Use Log Loss
```

SECTION C — The Full Picture: Regression vs. Classification

	Regression	Classification
Output type	Continuous number (142 mg/dL)	Discrete label (Diabetic / Not)
Primary metric	RMSE or MAE	Confusion Matrix → Precision/Recall/F1
"How wrong" question	By how many units was I off?	How many misclassified and which type?
Error asymmetry	Usually symmetric	Asymmetric — FP and FN have different costs
Baseline comparison	R^2 vs. mean-predictor baseline	Accuracy vs. majority-class baseline
Confidence metrics	N/A	Log Loss, AUC-ROC

SECTION D — Quick Revision: All Formulae in One Place

Regression:

$$\text{MAE} = \frac{1}{n}\sum|\hat{y} - y|$$

$$\text{MSE} = \frac{1}{n}\sum(\hat{y} - y)^2$$

$$\text{RMSE} = \sqrt{\text{MSE}}$$

$$R^2 = 1 - \frac{\sum(\hat{y}-y)^2}{\sum(\bar{y}-y)^2}$$

Classification:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP} \qquad \text{Recall} = \frac{TP}{TP + FN} \qquad \text{Specificity} = \frac{TN}{TN + FP}$$

$$\text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \qquad \text{FPR} = \frac{FP}{FP + TN}$$

$$\text{Log Loss} = -\frac{1}{n}\sum\left[y\log(p) + (1-y)\log(1-p)\right]$$

SECTION E — Real-World Domain Cheat Sheet

Domain	Task	Most Important Metric	Reason
Cancer/Disease Detection	Classification	**Recall**	Missing a real case (FN) is fatal
Spam Filter	Classification	**Precision**	Deleting real emails (FP) destroys trust
Fraud Detection	Classification	**Recall + F1**	Missing fraud (FN) means financial loss
House Price Prediction	Regression	**RMSE + R^2**	Large errors (outliers) matter a lot
Delivery Time Prediction	Regression	**MAE**	All errors roughly equal importance
Credit Risk Scoring	Classification	**AUC + Log Loss**	Need calibrated probabilities for risk bands
Weather Forecasting	Regression	**RMSE**	Extreme weather errors are costly
Resume Screening	Classification	**Precision**	False positives waste recruiter time
Fire/Flood Alarm	Classification	**Recall**	Missing a real event (FN) is catastrophic
Student Score Prediction	Regression	**MAE + R^2**	Interpretable, understand explanatory power

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> **The one-line philosophy behind all evaluation metrics:** *"A metric is just a formalised answer to the question — in your specific domain, what does it mean to be wrong, and how much does each type of wrong cost you?"*
